Duzhen Zhang

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PROFILE

Postdoctoral researcher in AI4BioMedicine at Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), working closely with Prof. Le Song. Research interests span LLM (Coding) Agents, Lifelong Agents, Agents4BioMedicine and Lifelong/Continual Learning in (Multimodal) LLMs. Published in ACL, EMNLP, NeurIPS, CVPR, ICML, SIGIR, ACM MM, AAAI, Npj Digital Medicine, IEEE TPAMI, IEEE TASLP, IEEE TMM, and other venues.

CURRENT POSITION

Postdoctoral Researcher in AI4BioMedicine
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) Sep 2024 – Present

- Adviser: Prof. Le Song
- Responsibilities: Develop and benchmark autonomous LLM agents for biomedical AI, enabling knowledge graph construction, multimodal ML coding, and end-to-end AI-driven discovery in BioMedicine.
- Focus:
  (1) LLM (Coding) Agents, Lifelong Agents, Agent Memory, Agents4BioMedicine;
  (2) Lifelong/Continual Learning in (Multimodal) LLMs;
  (3) Multimodal Large Language Models, Multimodal Learning

EDUCATION

Ph.D. in School of Artificial Intelligence
Institute of Automation, Chinese Academy of Sciences (CASIA) Sep 2019 – Jun 2024

- Adviser: Prof. Bo Xu (the director of CASIA), Prof. Tielin Zhang
- Focus:
  (1) Emotion Analysis in Conversational Systems;
  (2) Brain-inspired Intelligence, Spiking Neural Networks, Reinforcement Learning;
  (3) Lifelong/Continual Learning in Information Extraction;

B.Eng. in Software Engineering
Shandong University (SDU, Jinan) Sep 2015 – Jun 2019

PUBLICATIONS

ACCEPTED PAPERS
(*** denotes equal contribution and † denotes the corresponding author.**)

[1] **Duzhen Zhang***, Zixiao Wang*, Zhong-Zhi Li*, Yahan Yu, Shuncheng Jia, Jiahua Dong, Haotian Xu, Xing Wu, Yingying Zhang, Tielin Zhang, Jie Yang, Xiuying Chen†, and Le Song†, "*MedKGent: A Large Language Model Agent Framework for Constructing Temporally Evolving Medical Knowledge Graph*", Npj Digital Medicine (**Npj DM2026, Nature Family, IF: 18.0**).

[2] **Duzhen Zhang***, Zhongzhi Li*, Ming-Liang Zhang, Jiaxin Zhang, Zengyan Liu, Yuxuan Yao, Haotian Xu, Junhao Zheng, Xiuyi Chen, Yingying Zhang, Fei Yin, Jiahua Dong, Zhijiang Guo†, Le Song†, and Cheng-Lin Liu†, "*From System 1 to System 2: A Survey of Reasoning Large Language Models*", IEEE Transactions on Pattern Analysis and Machine Intelligence (**IEEE TPAMI'2025**).

[3] **Duzhen Zhang***†, Yong Ren*, Zhong-Zhi Li, Yahan Yu, Jiahua Dong, Chenxing Li, Zhilong Ji, and Jinfeng Bai, "*Enhancing Multimodal Continual Instruction Tuning with BranchLoRA*", in Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL'2025 Main**).

[4] **Duzhen Zhang**, Feilong Chen, and Xiuyi Chen†, "*DualGATs: Dual Graph Attention Networks for Emotion Recognition in Conversations*", in Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (**ACL'2023 Main**).

[5] **Duzhen Zhang***, Yahan Yu*, Jiahua Dong†, Chenxing Li, Dan Su, Chenhui Chu†, and Dong Yu, "*MM-LLMs: Recent Advances in MultiModal Large Language Models*", in Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (**ACL'2024 Findings**).

[6] **Duzhen Zhang***, Yahan Yu*, Qiaoyi Su, Jiahua Dong, and Tielin Zhang†, "*Progressive Multimodal Alignment for Continual Instruction Tuning*", in Proceedings of the 34th ACM International Conference on Multimedia (**ACM MM'2026**).

[7] **Duzhen Zhang***, Wei Cong*, Jiahua Dong†, Yahan Yu, Xiuyi Chen, Yonggang Zhang, and Zhen Fang, "*Continual Named Entity Recognition without Catastrophic Forgetting*", in Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (**EMNLP'2023 Main**).

[8] **Duzhen Zhang***, Yahan Yu*, Feilong Chen, and Xiuyi Chen†, "*Decomposing Logits Distillation for Incremental Named Entity Recognition*", in Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR'2023**).

[9] **Duzhen Zhang***, Hongliu Li*, Wei Cong, Rongtao Xu, Jiahua Dong, and Xiuyi Chen†, "*Task Relation Distillation and Prototypical Pseudo Label for Continual Named Entity Recognition*", in Proceedings of the 32nd ACM International Conference on Information and Knowledge Management (**CIKM'2023 Oral**).

[10] **Duzhen Zhang***, Tielin Zhang*†, Shuncheng Jia, and Bo Xu†, "*Multi-scale Dynamic Coding improved Spiking Actor Network for Reinforcement Learning*", in Proceedings of the 36th Association for the Advancement of Artificial Intelligence (**AAAI'2022 Oral**).

[11] **Duzhen Zhang**, Shuncheng Jia, and Qingyu Wang, "*Recent Advances and New Frontiers in Spiking Neural Networks*", in Proceedings of the 31st International Joint Conference on Artificial Intelligence (**IJCAI'2022**).

[12] Jiahua Dong*, **Duzhen Zhang***, Yang Cong, Wei Cong, Henghui Ding, and Dengxin Dai, "*Federated Incremental Semantic Segmentation*", the IEEE/CVF Conference on Computer Vision and Pattern Recognition 2023 (**CVPR'2023**).

[13] **Duzhen Zhang**, Chenxing Li†, Jiahua Dong, Qi Liu, and Dong Yu†, "*Exploring Stability-Plasticity Trade-offs for Continual Named Entity Recognition*", IEEE Transactions on Audio, Speech, and Language Processing (**IEEE TASLP'2025**).

[14] **Duzhen Zhang***, Yahan Yu*, Chenxing Li†, Jiahua Dong, Dong Yu†, "*Federated Incremental Named Entity Recognition*", IEEE Transactions on Audio, Speech, and Language Processing (**IEEE TASLP'2025**).

[15] **Duzhen Zhang***, Feilong Chen*, Jianlong Chang, Xiuyi Chen†, and Qi Tian, "*Structure Aware Multi-Graph Network for Multi-Modal Emotion Recognition in Conversations*", IEEE Transactions on MultiMedia (**IEEE TMM'2023**).

[16] **Duzhen Zhang***, Yahan Yu*, Xiuyi Chen, Chenxing Li, and Dong Yu, "*Type-Balanced Contextual Learning for Incremental Named Entity Recognition*", IEEE Transactions on Artificial Intelligence (**IEEE TAI'2026**).

[17] **Duzhen Zhang***, Yong Ren*, Chenxing Li, Dong Yu, and Tielin Zhang†, "*Information-Theoretic Complementary Prompts for Improved Continual Text Classification*", Neural Networks (**NN'2025**).

[18] **Duzhen Zhang***, Xiuyi Chen*, Shuang Xu, and Bo Xu, "*Knowledge Aware Emotion Recognition in Textual Conversations via Multi-Task Incremental Transformer*", in Proceedings of the 28th International Conference on Computational Linguistics (**COLING'2020 Oral**).

[19] **Duzhen Zhang**, Zhen Yang, Fandong Meng, Xiuyi Chen†, and Jie Zhou, "*TSAM: A Two-Stream Attention Model for Causal Emotion Entailment*", in Proceedings of the 29th International Conference On Computational Linguistics (**COLING'2022 Oral**).

[20] **Duzhen Zhang**, Tielin Zhang†, Shuncheng Jia, Qingyu Wang, and Bo Xu†, "*Tuning Synaptic Connections instead of Weights by Genetic Algorithm in Spiking Policy Network*", Machine Intelligence Research (**MIR'2024**).

[21] Loka Li, **Duzhen Zhang**†, Xingbo Du, Leonard Song, Zixiao Wang, Assanali Aukenov, Noel Thomas, Shakhnazar Sailaukan, Yonghan Yang, Feilong Chen, Jiahua Dong, Kun Zhang, Bin Zhang, and Le Song†, "*BioXArena: Benchmarking LLM Agents on Multi-Modal Biomedical Machine Learning Tasks*", in Proceedings of the 40th Annual Conference on Neural Information Processing Systems (**NeurIPS'2026**).

[22] Jiahua Dong*, Wenqi Liang*, Hongliu Li†, **Duzhen Zhang**†, Meng Cao, Henghui Ding, Salman Khan, and Fahad Khan, "*How to Continually Adapt Text-to-Image Diffusion Models for Flexible Customization?*", in Proceedings of the 38th Annual Conference on Neural Information Processing Systems (**NeurIPS'2024**).

[23] Yahan Yu, **Duzhen Zhang**†, Yong Ren, Xuanle Zhao, Xiuyi Chen, and Chenhui Chu, "*Progressive LoRA for Multimodal Continual Instruction Tuning*", in Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL'2025 Findings**).

[24] Yahan Yu, **Duzhen Zhang**†, Xiuyi Chen, and Chenhui Chu, "*Flexible Weight Tuning and Weight Fusion Strategies for Continual Named Entity Recognition*", in Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (**ACL'2024 Findings**).

[25] Feilong Chen*, **Duzhen Zhang***, Minglun Han, Xiuyi Chen, Jing Shi, Shuang Xu, and Bo Xu, "*VLP: A Survey on Vision-Language Pre-training*", Machine Intelligence Research (**MIR'2023**).

UNDER REVIEW

[25] **Duzhen Zhang***, Yong Ren*, Wei Cong, Junhao Zheng, Zhong-Zhi Li, Xuanle Zhao, Ye Bai, Feilong Chen, Qi Tian, and Tielin Zhang†, "*Revisiting Continual Semantic Segmentation with Pre-trained Vision Models*", IEEE Transactions on Pattern Analysis and Machine Intelligence (**IEEE TPAMI'2026**).

[26] **Duzhen Zhang***, Xingjian Du*, Yahan Yu, Qiaoyi Su, Ruibin Yuan, Zixiao Wang, Min Zeng, Shuncheng Jia, Jiahua Dong, and Tielin Zhang†, "*LoRARefine: Gradient-Guided LoRA Refinement for Multimodal Continual Instruction Tuning*", in Proceedings of the Forty-First AAAI Conference on Artificial Intelligence (**AAAI'2027**).

OTHER PUBLICATIONS

ACCEPTED PAPERS

[1] Min Zeng, Zaifu Zhan, Kai Yu, Chao Zeng, **Duzhen Zhang**, Shuang Zhou, Yu Hou, Mingquan Lin, Rui Zhang†, "*Large language models for digital mental health: an HCI-centered scoping review*", Npj Digital Medicine (**Npj DM2026**).

[2] Jiahua Dong*, Wenqi Liang*, Hongliu Li, Yang Cong†, **Duzhen Zhang**, Hanbin Zhao, Henghui Ding, Yulun Zhang, Salman Khan, and Fahad Shahbaz Khan, "*Crafting Your Evolving Dreams: Concept-Incremental Versatile Customization*", IEEE Transactions on Pattern Analysis and Machine Intelligence (**IEEE TPAMI'2026**).

[3] Junhao Zheng*, Chengming Shi*, Xidi Cai*, Qiuke Li*, **Duzhen Zhang**, Chenxing Li, Dong Yu, and Qianli Ma†, "*Lifelong Learning of Large Language Model based Agents: A Roadmap*", IEEE Transactions on Pattern Analysis and Machine Intelligence (**IEEE TPAMI'2025**).

[4] Xingbo Du, Loka Li, **Duzhen Zhang**, Leonard Song, and Le Song, "*Knowing What is Missing: Efficient Conversational Memory via Explicit Evidence-Gap Tracking*", in Proceedings of the 40th Annual Conference on Neural Information Processing Systems (**NeurIPS'2026**).

[5] Xuanle Zhao, Zilin Sang, Yuxuan Li, Qi Shi, Weilun Zhao, Shuo Wang, **Duzhen Zhang**, Xu Han, Zhiyuan Liu, and Maosong Sun, "*AutoReproduce: Automatic AI Experiment Reproduction with Paper Lineage*", in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL'2026 Main**).

[6] Zixiao Wang, **Duzhen Zhang**, Juntian Zhang, Yuhan Liu, Guoming Li, Haolun Wu, Le Song, and Xiuying Chen†, "*GlossaGen: Making Academic Translation Smarter with Glossing*", in Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (**ACL'2026 Findings**).

[7] Xuanle Zhao, Shuxin Zeng, Xinyuan Cai, Xiang Cheng, **Duzhen Zhang**, Xiuyi Chen, and Bo Xu, "*TinyChemVL: Advancing Chemical Vision-Language Models via Efficient Visual Token Reduction and Complex Reaction Tasks*", in Proceedings of the 40th Association for the Advancement of Artificial Intelligence (**AAAI'2026**).

[8] Jiahua Dong*, Xudong Wang*, Wenqi Liang, Zongyan Han, Meng Cao, **Duzhen Zhang**, Hanbin Zhao, Zhi Han, Salman Khan, and Fahad Shahbaz Khan, "*Bring Your Dreams to Life: Continual Text-to-Video Customization*", in Proceedings of the 40th Association for the Advancement of Artificial Intelligence (**AAAI'2026**).

[9] Zixiao Wang, **Duzhen Zhang**, Ishita Agarwal, Shen Gao, Le Song, and Xiuying Chen†, "*Beyond Profile: From Surface-Level Facts to Deep Persona Simulation in LLMs*", in Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (**ACL'2025 Findings**).

[10] Xuanle Zhao, **Duzhen Zhang**, Liyuan Han, Tielin Zhang, and Bo Xu, "*ODE-based Recurrent Model-free Reinforcement Learning for POMDPs*", in Proceedings of the 37th Annual Conference on Neural Information Processing Systems (**NeurIPS'2023**).

[11] Feilong Chen, **Duzhen Zhang**, Xiuyi Chen, Jing Shi, Shuang Xu, and Bo Xu, "*Unsupervised and Pseudo-Supervised Vision-Language Alignment in Visual Dialog*", in Proceedings of the 30th ACM International Conference on Multimedia (**ACM MM'2022**).

[12] Manjie Xu, Chenxing Li, **Duzhen Zhang**, Dan Su, Wei Liang, and Dong Yu, "*Prompt-guided Precise Audio Editing with Diffusion Models*", in Proceedings of the 41st International Conference on Machine Learning (**ICML'2024**).

[13] Qingyu Wang, Tielin Zhang, Minglun Han, Yi Wang, **Duzhen Zhang**, and Bo Xu, "*Complex Dynamic Neurons Improved Spiking Transformer Network for Efficient Automatic Speech Recognition "*, in Proceedings of the 37th Association for the Advancement of Artificial Intelligence (**AAAI'2023**).

INTERN EXPERIENCE

AI Lab, Tencent
Research Intern — Jan 2024 – Jul 2024.

- Mentor: Dr. Chenxing Li and Dr. Dong Yu.
- Worked on Continual Learning in Information Extraction and Multimodal LLM.
- The outcomes were submitted to ACL'24 and IEEE TASLP.

Search Strategy Department, Baidu
Research Intern — Apr 2023 – Jul 2023

- Mentor: Dr. Xiuyi Chen.
- Worked on Continual Learning in Information Extraction and LLM Inference Acceleration.
- The outcomes were accepted by SIGIR'23 and CIKM'23.

Pattern Recognition Center, Weixin Group (WXG), Tencent
Research Intern — Mar 2021 – Aug 2021

- Mentor: Dr. Zhen Yang and Dr. Fandong Meng.
- Worked on Emotion Anayasis in Conversations.
- The outcomes were accepted by COLING'22.

Natural Language Processing Center, Technology Group (TEG), Tencent
Research Engineering Intern — May 2020 – Aug 2020

- Mentor: Dr. Suncong Zheng.
- Worked on Neural Chinese Word Segmentation incorporating Prior Vocabulary Knowledge.
- By incorporating prior knowledge of the dictionary into the neural Chinese word segmentation model, problems related to OOV and cross-domain word segmentation have been alleviated.

PROFESSIONAL SERVICE

- Outstanding Area Chair: ARR 2025 Oct. (EACL 2026).
- Area Chair: ARR 2025 Oct.(EACL 2026), ARR 2026 Jan.(ACL 2026), ARR 2026 May(EMNLP 2026), ARR 2026 Aug.(NAACL 2027), AAAI 2027
- Conference Reviewer: AAAI 2023–2024, 2026; ACL 2023–2025; IJCAI 2023–2024; EMNLP 2023–2025; CVPR 2024–2026; NAACL 2025; ICCV 2025; NeurIPS 2025; ICLR 2026-2027; ECCV 2026; ICML 2026; ACM MM 2026, WWW2027.
- Journal Reviewer: Npj Digital Medicine, Npj Health Systems, IEEE TPAMI, IEEE TASLP, IEEE TMM.

AWARDS & SCHOLARSHIPS

- Second Prize Scholarship of Institute of Automation, Chinese Academy of Sciences — Dec 2022
- First Prize Scholarship for Outstanding Students of Shandong University — Sep 2017
- Second Prize of National College Student Mathematics Competition — Oct 2017

IT SKILLS

- Programming language: Python > C++ == C == Shell > Java
- Deep learning framework: Pytorch > Tensorflow

[*CV compiled on 2026-10-26*]